Saravanakumar Arunachalam

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Professional Summary

Backend-focused Software Engineer with 15+ years of experience designing and building highly available, reliable, and resilient distributed systems. Specialized in Golang, Java, and Python for backend platforms, APIs, and large-scale event-driven data pipelines. Proven expertise in designing architectures that ensure system stability, scalability, and observability. Strong hands-on experience with cloud-native technologies (AWS, Azure, Kubernetes), infrastructure automation (Terraform, Helm), and observability stacks (Grafana, Prometheus, Loki). Currently a Senior Platform Engineer at GEICO in Charlotte, building and leading end-to-end delivery of Kubernetes-native AI infrastructure — RAG pipelines, autoscaled LLM inference, and agentic AI patterns. Passionate about solving complex challenges in data-intensive environments while maintaining high performance and operational excellence.

Skills

AI/ML & GenAI: RAG Pipelines, LLM Inference & Serving, Agentic AI Patterns, Vector Databases & Embeddings, Prompt Engineering, Triton Inference Server, KEDA GPU Autoscaling, LLMOps, Model Deployment & Monitoring

Backend & Distributed Systems: Golang, Java, Python, Spring Boot, REST, gRPC, Microservices, System Design

Data & Streaming: Kafka, Kinesis, Pub/Sub, Flink, Event-Driven Architecture, High-Throughput Pipelines, Stream Processing, Real-time Analytics

Cloud & DevOps: AWS, Azure, Kubernetes, Docker, Terraform, Helm, Jenkins, CloudFormation, Knative, KEDA, Istio, Service Mesh

Databases & Storage: Postgres, MySQL, Oracle, MongoDB, DynamoDB, ElasticSearch, S3

Reliability & Observability: Grafana, Prometheus, Loki, Jaeger, SRE, Resilience Engineering, High Availability, Fault Tolerance, BCDR

Leadership & Delivery: Technical Leadership, Project Leadership, Team Mentoring, Cross-functional Collaboration, Stakeholder Management, Agile (Scrum/Kanban)

Architecture & Practices: Distributed Systems, CQRS, DDD, Scalability, TDD, CI/CD

Professional Experience

Senior Platform Engineer – Geico

Charlotte, NC | Jun 2024 – Present

- Leading end-to-end design, build, and delivery of AI platform initiatives — owning architecture, implementation, and production operations for RAG pipelines, autoscaled inference, and agentic AI workloads.
- Architected and developed a serverless platform with high availability, stability, and resilience on Kubernetes.
- Designed and implemented BCDR (Business Continuity and Disaster Recovery) strategies, including data replication and failover mechanisms.
- Integrated observability stack (Grafana, Prometheus, Loki) to ensure proactive monitoring and operational reliability.
- Delivered scalable CaaS/FaaS solutions leveraging Knative, KEDA, and Istio.
- Built Kubernetes-native infrastructure for production workloads, including Kafka pipelines, RAG pipelines, autoscaled inference, and agentic AI patterns.

Senior Software Engineer – Fidelity Investments (Contract)

Remote | Apr 2024 – Jun 2024

- Developed resilient backend services in Golang to support high-frequency trading and cryptocurrency workflows.
- Designed APIs with focus on reliability, stability, and secure handling of financial transactions.
- Adopted TDD and CI/CD practices to improve code quality and ensure consistent availability.

Backend Software Engineer – VMware Carbon Black

Remote | May 2022 – Apr 2024

- Engineered a high-throughput data pipeline for the alert management system, designed to process 50k records per second and multi-billion records daily.
- Optimized backend microservices in Golang and Java to ensure low-latency alerting and stable pipeline performance.
- Enhanced reliability and scalability of event-driven pipelines using Kafka.
- Fine-tuned system performance for large-scale security analytics workloads.

Senior Consultant – Lincoln Financial Group

Remote | Dec 2019 – May 2022

- Migrated legacy monoliths to resilient microservices on AWS with high availability and failover designs.
- Implemented CI/CD pipelines and automated infra provisioning with Terraform to improve reliability.
- Collaborated on API and data model design ensuring system stability and scalability.

Senior Software Engineer – J.B. Hunt Transport Services

Lowell, Arkansas | Mar 2018 – Dec 2019

- Built resilient backend microservices with Spring Boot, supporting fault tolerance and scalability.
- Developed distributed order processing pipelines to maintain reliability under peak traffic.
- Ensured service availability through automated deployments on AWS and Kubernetes.

Earlier Roles – Syntel, Photon Infotech, LTI, IndigoTx Software Technologies

Multiple | 2010 – 2018

- Contributed to backend engineering and microservices design focusing on system availability and reliability.
- Worked on distributed APIs, backend services, and cloud migration projects.
- Mentored engineers and enforced practices for resilience, observability, and operational stability.

Projects

rag-pipeline-k8s (Jun 2026) — End-to-end RAG pipeline on Kubernetes with KEDA autoscaled inference workloads

https://github.com/saravanastar/rag-pipeline-k8s

keda-triton-autoscaling (In Progress) — GPU inference autoscaling with KEDA + Triton on Kubernetes

https://github.com/saravanastar/llm-inference-k8s

Education & Certifications

Bachelor's Degree – Computer Science, Sudharsan Engineering College, 2005 - 2009

Nano Degree – Machine Learning, Udacity, 2019

- AWS Certified Solutions Architect – Associate